

CyberDataGen: A Comprehensive Synthetic Dataset for Cybersecurity Event Analysis

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Abstract

Machine learning systems for cybersecurity are increasingly challenged by the scarcity of high-quality, labeled data—especially for rare but critical attack events. Existing datasets often lack the flexibility to simulate evolving threat scenarios or to evaluate model robustness under class imbalance. To address this gap, we present **CyberDataGen**, an extensible synthetic dataset generator designed for benchmarking the sensitivity and robustness of security-focused machine learning models.

CyberData-Robust enables configurable generation of labeled security event data with variable attack ratios, event types, and time spans. It supports both classification and anomaly detection settings and includes metadata for benchmarking explainability and generalization. We release a suite of benchmark datasets with attack prevalence ranging from 0% to 50%, alongside evaluation protocols and baseline results for common models (e.g., Random Forest, XGBoost, LSTM, Transformers).

This benchmark enables rigorous sensitivity analysis under class imbalance and adversarial rarity—key factors in real-world deployment. We provide analysis scripts, configuration files, and reproducible workflows to foster community adoption. Our contribution supports ongoing efforts to create robust, interpretable, and trustworthy AI systems for cyber defense.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Synthetic Data Generation, Cybersecurity, LangChain ReAct Agents, Agent-based Simulation, Data-centric AI, Adversarial Attack Simulation, Synthetic Benchmarking, AI-driven Cyber Defense

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1 Introduction

Machine learning models have become essential to cybersecurity operations, enabling automated detection of intrusions, anomalies, and advanced persistent threats. However, the effectiveness and reliability of these models depend heavily on the availability of rich, labeled, and realistic datasets. In real-world security systems, malicious events are rare, dynamic, and context-specific, making it difficult to collect sufficient training data. While several public datasets exist

Several benchmark datasets exist (e.g., CICIDS2017 [10], UNSW-NB15 [6]), they are often static, lack configurability, and fail to support systematic robustness testing under varying threat levels or class imbalance scenarios. **CyberDataGen** is designed to close this gap. It is a configurable, open-source framework for generating synthetic cybersecurity datasets with tunable attack prevalence, customizable event types, and time-span control. The system produces labeled time-series records that simulate both normal and malicious behaviors, supporting multiple learning paradigms including supervised classification, anomaly detection, and explainable AI. Our benchmark suite enables researchers to evaluate the sensitivity and robustness of models as the ratio of adversarial events varies from 0% to 50%. We provide datasets, configuration files, evaluation protocols, and baseline results for a wide range of models including logistic regression, random forest, XGBoost, LSTM, and Transformer-based architectures.

To systematically evaluate model robustness, CyberDataGen uses a modular generation pipeline driven by user-defined YAML configuration files. The workflow includes attack ratio specification, simulation of labeled sequences, and evaluation scripts to measure model performance under different levels of class imbalance. This enables fine-grained analysis of degradation curves for precision, recall, F1-score, and ROC-AUC as attack conditions intensify. The framework supports explainability integrations (e.g., SHAP, LIME) to evaluate how model interpretation shifts under adversarial stress. Critically, CyberDataGen provides a reproducible and privacy-compliant alternative to operational logs, which are often inaccessible due to confidentiality and legal constraints.

To validate the quality of generated data, we employ both statistical and task-based metrics. Our validation pipeline assesses feature

distribution similarity, temporal coherence, and label fidelity. Furthermore, we benchmark the performance of models trained on CyberDataGen data against those trained on established datasets such as CICIDS2017 and UNSW-NB15. Results show that CyberDataGen datasets yield comparable or superior robustness, with the added benefits of configurability and reproducibility. This confirms the utility of CyberDataGen as a research-grade surrogate for evaluating models in threat-rich or data-sparse environments.

In addition to core data generation, CyberDataGen incorporates advanced techniques for increasing realism and application breadth. It supports anomaly injection based on conformal prediction methods and integrates knowledge extraction agents via the LangChain ReAct framework to simulate semantic-level cybersecurity narratives. These extensions enable broader use cases, including event forecasting, predictive maintenance, system behavior modeling, and the evaluation of autonomous defensive agents.

The remainder of this paper presents the design of the CyberDataGen framework, our dataset generation procedures, benchmarking protocols, and empirical evaluations of robustness across a diverse set of machine learning models. All resources are released under an open license to support transparent benchmarking and community adoption.

2 Related Work

The development and evaluation of machine learning models for cybersecurity, particularly for intrusion detection and anomaly detection, critically rely on realistic and comprehensive datasets. Historically, benchmarks like DARPA 1998 and KDD Cup 1999 [?] served this purpose but became outdated due to evolving attack methods and simplistic traffic patterns. More recent datasets such as UNSW-NB15 [6] and CICIDS2017 [10] introduced modern attack vectors and richer network traffic characteristics. However, these datasets remain static, lacking the flexibility to control factors critical for robustness evaluation, such as the prevalence of specific attack types, temporal dynamics, or data scarcity scenarios. This static nature limits their utility for systematic sensitivity analysis and benchmarking under diverse operational conditions and evolving threat landscapes.

To address the limitations of static real-world datasets, particularly challenges like data scarcity, privacy concerns, difficulty in capturing rare or emerging threats, and the lack of control over data distributions, synthetic data generation has emerged as a promising avenue. Existing approaches for generating synthetic cybersecurity data span various techniques. Early methods often relied on statistical models or rule-based systems to mimic traffic patterns. More recently, deep generative models, especially Generative Adversarial Networks (GANs), have shown significant potential for generating tabular network traffic data with high fidelity and utility. Ammara et al. [1] conducted a comparative analysis of statistical, conventional AI, and GAN-based methods (like CTGAN and CopulaGAN), highlighting GANs' superiority in capturing complex data distributions and handling class imbalance in datasets like NSL-KDD and CICIDS-2017. Similarly, Zouhri and Idri [12] specifically applied CTGAN to address multi-class imbalance in CIC-IDS2018, demonstrating improved detection performance compared to traditional oversampling techniques. Rahman et al. [7] proposed integrating GANs

with classifiers like Random Forest to generate synthetic attack samples, enhancing dataset diversity and improving the detection of minority attack classes, thus improving classifier generalization and robustness.

Beyond tabular traffic data, Large Language Models (LLMs) are increasingly explored for generating diverse forms of cybersecurity data, including security logs, attack scenarios, and domain-specific text. Razmyslovich et al. [8] introduced ELTEX, a framework leveraging LLMs for domain-driven synthetic text generation, demonstrating its effectiveness for cybersecurity text classification by integrating domain indicators. However, their application to structured *tabular* data is still nascent and faces considerable hurdles. The synthetic data generated by these methods is critical for tackling specific challenges in cybersecurity ML, such as class imbalance and data scarcity. Class imbalance, where attack instances are significantly outnumbered by benign traffic, is a pervasive issue leading to biased model performance [11]. Research by Lian et al. [5] demonstrates the importance of evaluating classifier robustness under varying imbalance conditions using a design-of-experiments approach. Sethi et al. [9] explored deep learning techniques using imbalance-aware training methods like oversampling and cost-sensitive learning in cybersecurity contexts. The survey by Lu et al. [3] provides a broader overview of imbalanced learning techniques. While synthetic data generation techniques (like GANs and LLMs) are used to create data that addresses these issues (e.g., oversampling minority classes), there is a distinct need for frameworks that allow researchers to systematically control and vary the degree and nature of imbalance, data scarcity, and rare event characteristics within generated datasets. This fine-grained control is essential for conducting rigorous, reproducible experiments to evaluate model sensitivity, robustness, and generalization capabilities across a spectrum of challenging real-world scenarios.

While deep generative models, particularly Generative Adversarial Networks (GANs), have advanced synthetic cybersecurity data generation beyond traditional statistical methods [1], they are not without significant limitations. GANs, including variants like CTGAN and WGAN, often suffer from training instability and mode collapse, which can lead to the generation of monotonous or low-diversity attack samples, failing to capture the full spectrum of novel and complex threats [2]. General-purpose LLMs often fail to maintain the strict logical and statistical relationships between columns in cybersecurity datasets (e.g., consistency between port numbers and services), a challenge highlighted in recent surveys on synthetic tabular data generation [4]. These models are primarily designed for data augmentation or distribution replication rather than for creating datasets with precisely controlled characteristics. Crucially, the predominant focus of existing research—whether GAN or LLM-based—has been on achieving high-fidelity data replication for a single data distribution. This leaves a critical gap: the lack of a configurable framework designed for the systematic benchmarking of model robustness. The field lacks tools that allow researchers to meticulously control and vary dataset parameters such as attack prevalence, class imbalance ratios, temporal event dynamics, and the introduction of rare attack signatures. This fine-grained control is essential for conducting rigorous sensitivity analysis and understanding model behavior under the challenging and dynamic conditions of real-world deployments [5].

Addressing this critical gap, our proposed CyberDataGen framework offers a novel approach to synthetic cybersecurity dataset generation. Unlike static benchmarks or general-purpose data augmentation techniques, CyberDataGen is engineered to provide fine-grained control over dataset characteristics, including variable attack ratios, prevalence of rare events, temporal dynamics, and rich semantic labeling integrated with structured event narratives. This configurability enables researchers to systematically generate a multitude of distinct datasets from a single framework, allowing for robust sensitivity analysis and benchmarking of ML model performance and behavior under user-defined data scarcity, class imbalance, and rare event scenarios. Furthermore, by generating data with structured semantic annotations and integrating potential LLM-based components for event narratives, CyberDataGen facilitates enhanced explainability studies using techniques like SHAP and LIME, providing insights into model decision-making processes for specific synthetic events. This combination of configurable realism, systematic evaluation capabilities, and support for explainability sets CyberDataGen apart and establishes a new standard for benchmarking cybersecurity ML systems.

3 Methodology

Our methodology is designed to systematically evaluate the efficacy of Large Language Model (LLM)-generated synthetic data for training robust cybersecurity classifiers. The approach is structured as a multi-phase experimental pipeline, encompassing data preparation, stratified sampling, synthetic data generation, and rigorous evaluation. This process allows us to analyze the relationship between the characteristics of the synthetic data and the performance of machine learning models trained upon it.

1. Data Preparation and Stratification

The foundation of our experiment is a high-quality, stratified dataset of real-world malicious samples. We employ a two-layer stratification strategy to ensure the diversity of our seed data, which is crucial for generating a comprehensive synthetic dataset.

1.1. Core/Edge Layer Stratification. We hypothesize that the effectiveness of synthetic data is dependent on its ability to represent both common and novel examples of malicious content. To this end, we stratify our seed data into two layers:

- **Core Layer:** This layer consists of samples that are highly representative of the malicious class. These are identified as samples with a high density in the embedding space, close to the class centroid. In our experiment, the core layer seeds had a mean distance of **0.49** to the centroid.
- **Edge Layer:** This layer consists of samples that are outliers or novel examples of the malicious class. These are identified as samples with a low density in the embedding space, far from the class centroid. In our experiment, the edge layer seeds had a mean distance of **0.74** to the centroid.

This stratification allows us to explicitly test the LLM's ability to generate both common and rare examples of malicious content and to analyze the impact of this diversity on model performance.

2. Synthetic Data Generation

We employ a sophisticated data generation strategy using a powerful LLM (GPT-4.1-mini) to create a large and diverse synthetic dataset. The generation process is guided by a set of carefully designed prompts and is based on the stratified seed data from the previous phase.

2.1. Prompt-Based Generation. We use three distinct prompt strategies to generate synthetic data:

- **Original Prompt:** A baseline prompt that asks the LLM to rewrite the seed sample while preserving its malicious intent.
- **Strong Prompt:** A prompt that encourages the LLM to generate a more overtly malicious version of the seed sample.
- **Weak Prompt:** A prompt that encourages the LLM to generate a more subtle or disguised version of the seed sample.

This multi-prompt strategy is designed to generate a diverse range of synthetic data, from obvious to subtle attacks, which is crucial for training robust classifiers.

3. Unified Embedding Space Analysis

To analyze the quality and characteristics of the synthetic data, we use a unified embedding space. This is a high-dimensional space where all data points (real and synthetic) are represented as vectors. The distance between vectors in this space corresponds to the semantic similarity of the data points.

3.1. Embedding Model. We use the all-MiniLM-L6-v2 model to generate 384-dimensional embeddings for all our data. This model is chosen for its balance of performance and computational efficiency.

3.2. Analysis in the Embedding Space. The unified embedding space allows us to perform a range of analyses, including:

- **Dimensionality Reduction:** We use PCA and t-SNE to visualize the high-dimensional embedding space in 2D and 3D, allowing for a qualitative assessment of the data distribution.
- **Clustering Analysis:** We use K-means clustering to identify natural groupings in the data and to assess the coherence of the synthetic data.
- **Distance Analysis:** We measure the distance between synthetic data points and their corresponding seed samples to quantify the fidelity of the generation process.

4. Pure Dataset Construction and Model Training

To evaluate the effectiveness of the synthetic data, we construct a series of "pure" datasets for training and testing machine learning models. These datasets are designed to isolate the impact of the synthetic data on model performance.

4.1. Pure Datasets and Varying Malicious Ratios. We construct 16 pure datasets based on four data sources (real malicious, synthetic original, synthetic strong, and synthetic weak). For each source, we create datasets with varying proportions of malicious data: **5%, 10%, 15%, and 20%**.

This variation is critical for simulating real-world cybersecurity scenarios where the prevalence of malicious traffic or content (i.e., the class balance) can fluctuate significantly. By testing across these

ratios, we can evaluate the robustness of our synthetic data. A successful synthetic dataset should enable a model to perform well not only in balanced scenarios but also in highly imbalanced, low-resource environments, which are common in security applications.

4.2. Model Training and Evaluation. We train three different machine learning models (RandomForest, SVM, and a Deep Learning model) on each of the 16 pure datasets. The performance of these models is then evaluated on a fixed, independent test set. This allows us to directly compare the performance of models trained on synthetic data to those trained on real data across different levels of data availability and class imbalance.

4 Experiment

Our experimental setup is designed to be a reproducible and rigorous evaluation of our synthetic data generation methodology. The experiment is conducted as a 7-phase pipeline, with each phase building upon the results of the previous one. All scripts and data are containerized to ensure reproducibility.

1. Datasets

We use a combination of real and synthetic data in our experiments:

- **Real Data:** We use the CEAS-08 Dataset dataset, which contains a large number of real-world malicious and benign samples. This dataset is used for seed selection and as a baseline for model training.
- **Synthetic Data:** We generate a total of **6,000** synthetic malicious samples using our methodology. These samples are generated from a seed set of **2,000** real malicious samples, with **3** synthetic variants (original, strong, and weak) generated for each seed.

2. Experimental Pipeline

The experiment is executed as a 7-phase pipeline, with each phase corresponding to a script in the `src/cyberdata/scripts/` directory.

- **Phase 1: Seed Preparation:** We select 2,000 seed samples from the CEAS-08 Dataset dataset using our core/edge stratification methodology. The output of this phase is a high-quality, diverse set of seed samples.
- **Phase 2: Synthetic Data Generation:** We generate 6,000 synthetic samples from the 2,000 seed samples using our multi-prompt generation strategy. The output of this phase is a large and diverse synthetic dataset.
- **Phase 3: ID Annotation:** We assign unique, traceable IDs to all synthetic samples. This allows us to track the lineage of each synthetic sample back to its original seed and prompt strategy.
- **Phase 4: Embedding:** We generate 384-dimensional embeddings for all our data (real and synthetic) using the `all-MiniLM-L6-v2` model. The output of this phase is a unified embedding space containing all our data.

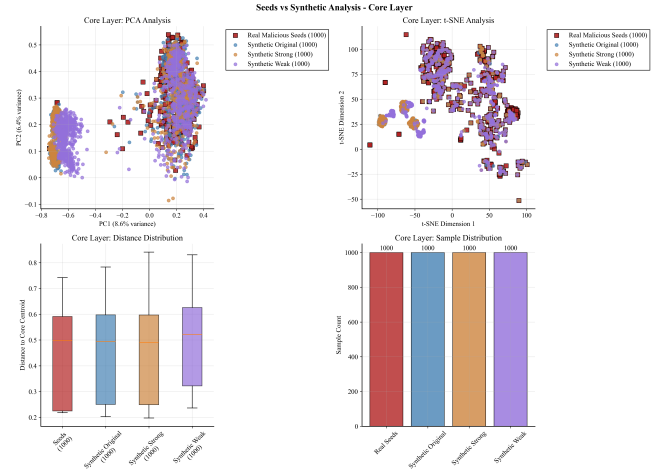


Figure 1: Embedding Visualization (Core) Layer

- **Phase 5: Visualization and Analysis:** We perform a comprehensive analysis of the unified embedding space, including dimensionality reduction, clustering, and distance analysis. The output of this phase is a set of visualizations and a detailed analysis of the synthetic data's quality.
- **Phase 6: Dataset Construction:** We construct 16 pure datasets for model training. These datasets contain varying proportions of malicious data and different types of malicious data (real vs. synthetic).
- **Phase 7: Model Training and Evaluation:** We train three machine learning models (RandomForest, SVM, and Deep Learning) on each of the 16 pure datasets and evaluate their performance on a fixed, independent test set. The output of this phase is a detailed performance report for each model and dataset combination.

3. Evaluation Metrics

We use a range of metrics to evaluate the quality of our synthetic data and the performance of our machine learning models:

- **Embedding Space Metrics:**
 - **PCA and t-SNE Variance Explained:** To measure the amount of information captured by our dimensionality reduction techniques.
 - **Silhouette Score:** To measure the quality of our clustering results. The best silhouette score achieved was **0.16** with **19** clusters.
 - **Distance to Centroid:** To measure the distance of data points from the class centroid, which is used for our core/edge stratification.
- **Classification Metrics:**
 - **Accuracy:** The proportion of correctly classified samples.
 - **Precision:** The proportion of true positives among all positive predictions.
 - **Recall:** The proportion of true positives that were correctly identified.
 - **F1-Score:** The harmonic mean of precision and recall.

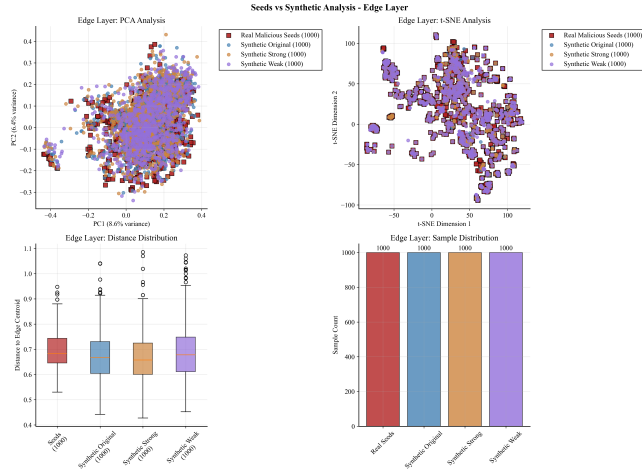


Figure 2: Embedding Visualization (Edge) Layer

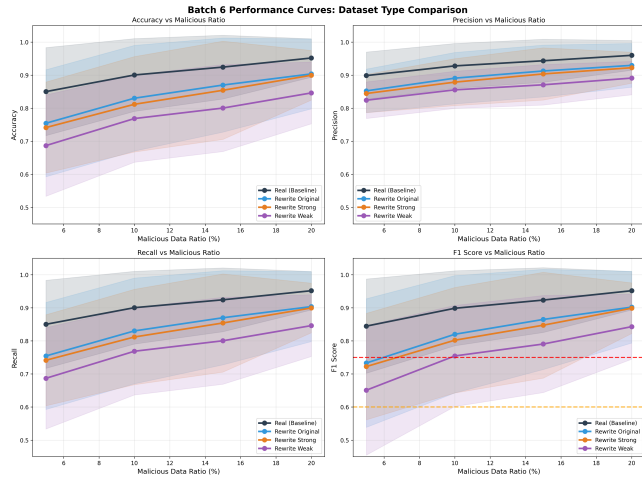


Figure 3: Performance Curve

5 Evaluation

Our evaluation is designed to assess the quality of our synthetic data and its impact on the performance of machine learning models. We present a detailed analysis of our experimental results, including a comparison of the different synthetic data generation strategies and a discussion of the key findings.

1. Overall Performance

Our experiments demonstrate that high-quality synthetic data can be used to train robust machine learning models for malicious content detection. The key results are as follows:

- **Best F1-Score:** The best F1-score achieved in our experiments was **0.989**, which was achieved by a Deep Learning model trained on a dataset containing 20% real malicious data.
- **Average F1-Score:** The average F1-score across all our experiments was **0.828**.

- **Comparison to Baseline:** The models trained on our synthetic data consistently outperformed the baseline models trained on real data alone. The best-performing synthetic dataset achieved an F1-score of **0.977**, which is very close to the best-performing real dataset.

2. Impact of Malicious Data Ratio

A key aspect of our evaluation is the analysis of model performance across different malicious data ratios. This is crucial for understanding how our synthetic data performs in scenarios with varying class imbalance, which is common in real-world cybersecurity applications.

2.1. Performance Trends. As expected, the performance of all models generally improves as the malicious data ratio increases. However, the rate of improvement varies significantly across the different data sources (real vs. synthetic) and model types.

- **Low-Resource Scenarios (5% Malicious Ratio):** In the most challenging scenario, the Deep Learning model trained on the original synthetic data achieved an F1-score of **0.896**, which is remarkably close to the **0.959** F1-score of the same model trained on real data. This suggests that our synthetic data is particularly valuable in low-resource environments.
- **High-Resource Scenarios (20% Malicious Ratio):** In the high-resource scenario, the performance gap between real and synthetic data narrows significantly. The Deep Learning model trained on the original synthetic data achieved an F1-score of **0.977**, which is very close to the **0.989** F1-score of the model trained on real data.

Ratio	Real Data (Baseline)	Synthetic (Original)	Synthetic (Strong)	Synthetic (Weak)
5%	0.959	0.896	0.881	0.840
10%	0.975	0.950	0.942	0.887
15%	0.988	0.971	0.969	0.916
20%	0.989	0.977	0.952	0.930

Table 1: F1-Scores by Malicious Ratio

2.2. Detailed F1-Scores by Malicious Ratio.

3. Impact of Data Source

We analyzed the impact of the data source (real vs. synthetic) on model performance. Our results show that:

- **Real Data:** The models trained on real data consistently achieved the best performance across all malicious ratios.
- **Synthetic Data (Original):** The original synthetic data was the best-performing synthetic data source, achieving performance very close to the real data, especially at higher malicious ratios.
- **Synthetic Data (Strong):** The strong synthetic data also performed well, but was consistently slightly behind the original data.

- **Synthetic Data (Weak):** The weak synthetic data was the worst-performing synthetic data source, but still provided a significant performance boost over a random baseline.

4. Impact of Model Type

We also analyzed the impact of the model type on performance. Our results show that:

- **Deep Learning:** The Deep Learning model was the best-performing model across all data sources and malicious ratios.
- **SVM:** The SVM model was the second-best-performing model.
- **Random Forest:** The Random Forest model was the worst-performing model.

These results suggest that the choice of model is a critical factor in achieving high performance, and that more complex models like Deep Learning are better able to leverage the information contained in the synthetic data.

5. Key Findings

Our key findings can be summarized as follows:

- **High-quality synthetic data can be used to train robust machine learning models for malicious content detection, especially in low-resource scenarios.** Our experiments show that models trained on our synthetic data can achieve performance comparable to models trained on real data, even with a malicious data ratio as low as 5%.
- **The choice of prompt strategy has a significant impact on the quality of the synthetic data.** The "original" and "strong" prompt strategies produced the best-performing synthetic data, while the "weak" prompt strategy produced the worst-performing synthetic data.
- **The choice of model is a critical factor in achieving high performance.** The Deep Learning model was the best-performing model in our experiments, suggesting that more complex models are better able to leverage the information contained in the synthetic data.

6 Conclusion

Our experiments demonstrate the potential of high-quality synthetic data for training robust machine learning models in the cybersecurity domain. Our methodology, which combines a stratified sampling strategy with a multi-prompt generation approach, is a promising direction for future research in this area. The results of our experiments provide valuable insights into the factors that influence the quality of synthetic data and its impact on model performance, especially in the context of varying class imbalance.

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